MODULE 20 · SECTION 3 - TECHNIQUES TO IMPROVE GEN AI MODEL OUTPUT

Prompt engineering: zero-shot to ReAct

How to brief a model well, the named techniques the exam tests, and when to stop prompting and change something else.

SECTION 3 of 4 EXAM WEIGHT ~20% MODULE 20 of 33

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A prompt is a brief to a brilliant, very literal new colleague. **Vague in, vague out.**

Six parts of a strong prompt

PART	AT WRENFIELD
Role	"You are Wrenfield's senior copywriter"
Task	"Write a product description"
Context	The product facts and the audience
Constraints	"Under 120 words; never mention price"
Format	Headline, two short paragraphs, three bullets
Examples	Two approved descriptions to copy

REMEMBER

Google advises putting the role, key constraints and output format first, or in the system instructions.

A weak prompt, and a strong one

WEAK

"Write about this sofa."

- No audience or tone
- No facts supplied
- No length or format
- The model guesses everything

STRONG

A full brief

- Role: Wrenfield copywriter
- Facts: size, fabric, colours
- Constraints: 120 words, no price
- Format plus two examples

THE TAKEAWAY

Most poor output comes from a poor brief. Fix the prompt before reaching for tuning or a bigger model.

Zero-shot, one-shot, few-shot

TECHNIQUE	WHAT THE PROMPT CONTAINS
Zero-shot	Instructions only, no examples
One-shot	Instructions plus one example
Few-shot	Instructions plus several examples

THE TIEBREAKER

"Shot" means example. Count the examples in the prompt, and you have the name.

Few-shot: show, don't just tell

- **Google's advice:** include few-shot examples in your prompts.
- **Examples teach** format, tone, scope and judgement at once.
- **Keep their structure** identical, so the pattern is clear.
- **Use varied examples,** not five that look the same.
- **Too many** can make the model copy them too closely.

> THE SIGNAL
>
> Output is inconsistent in format or tone, and instructions alone have not fixed it: add few-shot examples.

Role prompting and system instructions

- **Role prompting** tells the model who to be: an expert, a persona, a voice.
- **It shapes** vocabulary, tone, depth and point of view.
- **System instructions** hold the role and rules for a whole conversation.
- **Users then chat** without repeating the brief each time.
- **A Gem** is a saved set of system instructions, from module 11.

THE SIGNAL

"Act as a..." or a fixed persona and tone across a whole conversation: role prompting, in system instructions.

Prompt chaining: one step at a time

01
Extract
Pull the facts from the supplier email.

›

02
Classify
Rate the delay: minor or serious.

›

03
Draft
Write the customer update.

›

04
Check
Review tone and accuracy.

THE TIEBREAKER

A big task done badly in one prompt: split it into a chain. Each step is simpler, testable and easier to fix.

Chain-of-thought: show the working

- **Ask the model** to reason step by step before answering.
- **Helps with** multi-step logic, rules and calculations.
- **Makes the reasoning** visible, so people can check it.
- **Thinking models** such as Gemini now reason internally by default.
- **For heavy arithmetic,** code execution is more reliable still.

THE SIGNAL

Errors on multi-step reasoning or calculations: chain-of-thought. It is the guide's term even though modern models often think by default.

Chain-of-thought or ReAct?

CHAIN-OF-THOUGHT

Thinks, then answers

- Reasoning only
- Uses what is in the prompt
- No tools, no lookups
- A pupil showing their working

REACT

Thinks, acts, observes, repeats

- Reasoning plus actions
- Calls tools and checks results
- Adapts to what it finds
- The agent loop from module 16

THE TIEBREAKER

Reasoning alone: chain-of-thought. Reasoning that interleaves with tool calls and observations: ReAct.

Phrases that point to each technique

THE QUESTION SAYS	CHOOSE
"No examples, just the instruction"	Zero-shot
"A single example of the format"	One-shot
"Several examples to match the style"	Few-shot
"Act as an expert adviser"	Role prompting
"Break the task into a sequence"	Prompt chaining
"Reason step by step"	Chain-of-thought
"Reason, call tools, observe, repeat"	ReAct

ON THE EXAM

Seven names, seven phrases. Most technique questions are decided by one phrase in the stem.

When prompting is not the fix

SYMPTOM AFTER GOOD PROMPTING	NEXT FIX
Facts missing or out of date	Ground it, or use RAG
A style prompts cannot hold	Fine-tune with examples
High-stakes or regulated output	Human in the loop
Wrong job for gen AI	Rules or traditional ML

REMEMBER

Prompting shapes how a model uses what it knows. It cannot give the model facts it does not have.

> Wrenfield's product descriptions come back in different layouts and tones each time.
>
> The instructions already describe the layout, and the team has many approved descriptions to copy.

What should they try first?

A Add two or three approved descriptions as examples

B Remove the instructions to keep the prompt short

C Raise the temperature

D Fine-tune a model straight away

◆ PAUSE AND DECIDE

What should they try first?

A Add two or three approved descriptions as examples

Few-shot examples teach format and tone together.

B Remove the instructions to keep the prompt short

Less guidance gives less consistency.

C Raise the temperature

More randomness, less consistency.

D Fine-tune a model straight away

Possible later, but skips the cheap fix.

THE SIGNAL

Inconsistent format or tone, examples available: few-shot.

> Wrenfield's support assistant sounds stiff and generic.
>
> The team adds: "You are Wrenfield's senior customer care adviser. You are warm, plain-spoken and brief."

Which technique is this?

A Chain-of-thought

B Few-shot prompting

C Role prompting

D ReAct

◆ PAUSE AND DECIDE

Which technique is this?

A Chain-of-thought

Asks for step-by-step reasoning; no persona.

B Few-shot prompting

Would add examples; this adds a persona.

C Role prompting

Sets who the model is, and how it sounds.

D ReAct

Interleaves reasoning with tool calls.

THE SIGNAL

A persona and voice for the model: role prompting.

> One long prompt asks the model to read a supplier email, extract the delay facts, rate the severity, draft a customer update, and check its tone.
>
> The results are muddled, and nobody can tell which step went wrong.

What should the team do?

A Make the single prompt even longer and more detailed

B Split it into a chain of prompts, one step each

C Lower the temperature

D Remove the tone check

◆ PAUSE AND DECIDE

What should the team do?

A Make the single prompt even longer and more detailed

More text in one prompt adds to the muddle.

B Split it into a chain of prompts, one step each

Each step is simpler, and testable on its own.

C Lower the temperature

Steadier wording; the steps stay tangled.

D Remove the tone check

Drops a requirement instead of fixing it.

THE SIGNAL

A complex task done badly in one go: prompt chaining.

> Wrenfield's assistant calculates delivery charges for multi-item orders, applying weight bands, a free-delivery threshold and a two-person surcharge.
>
> It often jumps straight to a wrong total.

Which prompting technique helps most?

A Role prompting

B One-shot prompting

C A shorter prompt

D Chain-of-thought: reason step by step

◆ PAUSE AND DECIDE

Which prompting technique helps most?

A Role prompting

A persona does not fix multi-step logic.

B One-shot prompting

One example helps format, not the reasoning.

C A shorter prompt

Less guidance for a hard problem.

D Chain-of-thought: reason step by step

Works through each rule before the total.

THE SIGNAL

Errors in multi-step reasoning: chain-of-thought.

> Wrenfield's returns agent must look up an order, read the result, check the courier's slots, read that result, and then decide what to offer.
>
> Each step depends on what the previous lookup returned.

Which approach fits best?

A Chain-of-thought

B Few-shot prompting

C ReAct: reason, act with tools, observe, repeat

D A fixed prompt chain written in advance

◆ PAUSE AND DECIDE

Which approach fits best?

A Chain-of-thought
Reasoning only; it cannot look anything up.

B Few-shot prompting
Examples do not perform lookups.

C ReAct: reason, act with tools, observe, repeat
Reasoning interleaved with tool calls and results.

D A fixed prompt chain written in advance
Fixed steps cannot adapt to what each lookup returns.

THE SIGNAL

Reasoning that alternates with tool calls: ReAct.

> A store manager wants Gemini to summarise each long head office email in two sentences. The results are already good with a plain instruction.

Which technique is she using, and is it enough?

A Zero-shot, and it is enough for this simple task

B Few-shot, and she should add twenty examples

C Fine-tuning, which she needs to add

D ReAct, because summaries need tools

◆ PAUSE AND DECIDE

Which technique is she using, and is it enough?

A Zero-shot, and it is enough for this simple task
Instruction only, and the results are already good.

B Few-shot, and she should add twenty examples
Extra effort for no gain on a simple task.

C Fine-tuning, which she needs to add
Heavy and unnecessary.

D ReAct, because summaries need tools
No tools are involved in summarising.

THE SIGNAL

A simple task done well with no examples: zero-shot is enough.

> After weeks of prompt work, Wrenfield's sales assistant still quotes old prices.
>
> Prices change daily, and the prompt already says "use current prices only".

What is the right next step?

A Add more examples of correct answers

B Strengthen the role and the tone

C Ask for step-by-step reasoning

D Ground answers in the live pricing data

◆ PAUSE AND DECIDE

What is the right next step?

A Add more examples of correct answers

Examples cannot supply today's prices.

B Strengthen the role and the tone

A persona does not know prices either.

C Ask for step-by-step reasoning

Reasoning about missing facts still fails.

D Ground answers in the live pricing data

Brings in today's prices at answer time.

THE TAKEAWAY

Missing or changing facts: prompting has hit its ceiling; ground it.

> Wrenfield plans to introduce a new "care card" layout for product pages.
>
> So far there is exactly one approved care card, which the team pastes into the prompt with the instructions.

Which technique is this?

A Zero-shot

B One-shot

C Few-shot

D Role prompting

◆ PAUSE AND DECIDE

Which technique is this?

A Zero-shot

Zero-shot uses no examples.

B One-shot

Instructions plus a single example.

C Few-shot

Few-shot needs several examples.

D Role prompting

No persona is being set.

THE SIGNAL

Count the examples: one example is one-shot.

Module 20 in five lines

- A strong prompt: role, task, context, constraints, format and examples.
- Zero, one or few-shot: count the examples. Few-shot fixes inconsistency.
- Role prompting sets a persona; chaining splits a big task into steps.
- Chain-of-thought reasons step by step; ReAct reasons and uses tools.
- Prompting cannot supply missing facts: then ground, tune or add people.